VIT-AP UNIVERSITY, ANDHRA PRADESH

Lab Sheet 7: MongoDB with Fronend

Branch/ Class: B.Tech/M.Tech **Date: 05-03-2025**
Faculty Name: Prof. S.Gopikrishnan **School:** SCOPE
Student name: Ponnoju Sreeram Siddardha **Reg. no.:** 23BCE9304

Question 1: Student Notes Manager (CRUD Operations)
Develop a simple web application that allows students to create, view, update, and delete study notes stored in MongoDB.

Modules

1. Add Note
2. View Notes
3. Edit Note
4. Delete Note

Functional Requirements

1. Add Note

Student enters:

- Title
- Subject
- Description

AJAX sends JSON data to server.
Example JSON request:

```
{
   "title":"MongoDB Basics",
   "subject":"Database",
   "description":"Introduction to MongoDB concepts"
   }
```

MongoDB stores document in **notes collection**.
Example MongoDB document:

```
{
   "_id":"ObjectId",
   "title":"MongoDB Basics",
   "subject":"Database",
   "description":"Introduction to MongoDB concepts",
   "created_date":"2026-03-05"
   }
```

2. View Notes

Display all notes stored in database.
AJAX request:
GET /notes
Server returns JSON:

```
[
   {
   "title":"MongoDB Basics",
   "subject":"Database",
```

```
"description":"Introduction to MongoDB concepts"
}
]
```

Frontend dynamically renders notes using JavaScript.

3. Update Note

Student can modify title or description.

AJAX request:

PUT /notes/{id}

```
{
"title":"MongoDB Advanced",
"description":"Aggregation and indexing"
}
```

MongoDB operation:

```
db.notes.updateOne(
{_id:ObjectId(id)},
{$set:{title:"MongoDB Advanced",description:"Aggregation and indexing"}}
)
```

4. Delete Note

AJAX request:

DELETE /notes/{id}

MongoDB query:

```
db.notes.deleteOne({_id:ObjectId(id)})
```

VIT-AP UNIVERSITY, ANDHRA PRADESH

Lab Sheet 7: MongoDB with Fronend

Branch/ Class: B.Tech/M.Tech **Date: 05-03-2025**
Faculty Name: Prof. S.Gopikrishnan **School:** SCOPE
Student name: Ponnoju Sreeram Siddardha **Reg. no.:** 23BCE9304

```
test> use student_notes
switched to db student_notes
student_notes> db.notes.insertOne({
| title: "MongoDB Basics",
| subject: "Database",
| description: "Introduction to MongoDB concepts",
| created_date: "2026-03-05"
| })
{
  acknowledged: true,
  insertedId: ObjectId('69a9aed5f56558da717c2907')
}
student_notes> db.notes.find()
[
  {
    _id: ObjectId('69a9aed5f56558da717c2907'),
    title: 'MongoDB Basics',
    subject: 'Database',
    description: 'Introduction to MongoDB concepts',
    created_date: '2026-03-05'
  }
]
student_notes> db.notes.find().pretty()
[
  {
    _id: ObjectId('69a9aed5f56558da717c2907'),
    title: 'MongoDB Basics',
    subject: 'Database',
    description: 'Introduction to MongoDB concepts',
    created_date: '2026-03-05'
  }
]
```

```
student_notes> db.notes.updateOne(
| { title: "MongoDB Basics" },
| { $set: { description: "Basic concepts of MongoDB database" } }
| )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
student_notes> db.notes.deleteOne(
| { title: "MongoDB Basics" }
| )
{ acknowledged: true, deletedCount: 1 }
student_notes> show collections
notes
student_notes> show dbs
admin           40.00 KiB
animal          80.00 KiB
config          72.00 KiB
local           64.00 KiB
student_notes   40.00 KiB
vehicles       112.00 KiB
student_notes>
```

VIT-AP UNIVERSITY, ANDHRA PRADESH

Lab Sheet 7: MongoDB with Fronend

Branch/ Class: B.Tech/M.Tech **Date: 05-03-2025**
Faculty Name: Prof. S.Gopikrishnan **School:** SCOPE
Student name: Ponnoju Sreeram Siddardha **Reg. no.:** 23BCE9304

```
student_notes> db.notes.updateOne(
| { _id: ObjectId("67c86f9c5c4b2a8e1c9a1234") },
| {
| $set:{
| title:"MongoDB Advanced",
| description:"Aggregation and indexing"
| }
| }
| )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 0,
  modifiedCount: 0,
  upsertedCount: 0
}
student_notes>
```

VIT-AP UNIVERSITY, ANDHRA PRADESH

Lab Sheet 7: MongoDB with Fronend

Branch/ Class: B.Tech/M.Tech **Date: 05-03-2025**
Faculty Name: Prof. S.Gopikrishnan **School:** SCOPE
Student name: Ponnoju Sreeram Siddardha **Reg. no.:** 23BCE9304

Question 2: Online Book Finder (Retrieval & Query Operations)
Develop a system where users can search, filter, and retrieve books using different MongoDB queries.

Modules

1. Search Books by Title
2. Filter Books by Category
3. Sort Books by Price or Rating
4. Display Top Rated Books
5. Pagination (Load More)

Database Structure
Collection: **books**
Example document:

```
{
  "_id":"ObjectId",
  "title":"JavaScript Essentials",
  "author":"John Smith",
  "category":"Programming",
  "price":450,
  "rating":4.5,
  "year":2023
  }
```

Functional Requirements

1. Search Books by Title
AJAX request:
GET /books/search?title=javascript
MongoDB query:
db.books.find({title:{$regex:"javascript",$options:"i"}})
Returns JSON results.

2. Filter Books by Category
AJAX request:
GET /books/category/programming
MongoDB query:
db.books.find({category:"Programming"})

3. Sort Books
Sort by price.
AJAX request:
GET /books/sort/price
MongoDB query:
db.books.find().sort({price:1})

VIT-AP UNIVERSITY, ANDHRA PRADESH

Lab Sheet 7: MongoDB with Fronend

Branch/ Class: B.Tech/M.Tech **Date: 05-03-2025**
Faculty Name: Prof. S.Gopikrishnan **School:** SCOPE
Student name: Ponnoju Sreeram Siddardha **Reg. no.:** 23BCE9304

Sort by rating.
db.books.find().sort({rating:-1})

4. Top Rated Books

AJAX request:
GET /books/top
MongoDB query:
db.books.find({rating:{$gte:4}}).limit(5)

5. Pagination

AJAX request:
GET /books?page=2
MongoDB query:

```
db.books.find()
    .skip(5)
    .limit(5)
```

VIT-AP UNIVERSITY, ANDHRA PRADESH

Lab Sheet 7: MongoDB with Fronend

Branch/ Class: B.Tech/M.Tech **Date: 05-03-2025**
Faculty Name: Prof. S.Gopikrishnan **School:** SCOPE
Student name: Ponnoju Sreeram Siddardha **Reg. no.:** 23BCE9304

```
  {
    _id: ObjectId('69a9b2020beeaeed997c290a'),
    title: 'Machine Learning Intro',
    author: 'Robert White',
    category: 'AI',
    price: 600,
    rating: 4.8,
    year: 2023
  },
  {
    _id: ObjectId('69a9b2020beeaeed997c290b'),
    title: 'HTML & CSS',
    author: 'Chris Green',
    category: 'Web Development',
    price: 300,
    rating: 3.9,
    year: 2021
  }
]
online_book_finder> db.books.find({
| title:{$regex:"javascript",$options:"i"}
| })
[
  {
    _id: ObjectId('69a9b2020beeaeed997c2907'),
    title: 'JavaScript Essentials',
    author: 'John Smith',
    category: 'Programming',
    price: 450,
    rating: 4.5,
    year: 2023
  }
]
online_book_finder> db.books.find({
| category:"Programming"
| })
[
  {
    _id: ObjectId('69a9b2020beeaeed997c2907'),
```

VIT-AP UNIVERSITY, ANDHRA PRADESH

Lab Sheet 7: MongoDB with Fronend

Branch/ Class: B.Tech/M.Tech **Date: 05-03-2025**
Faculty Name: Prof. S.Gopikrishnan **School:** SCOPE
Student name: Ponnoju Sreeram Siddardha **Reg. no.:** 23BCE9304

```
online_book_finder> db.books.find().sort({rating:-1})
[
  {
    _id: ObjectId('69a9b2020beeaeed997c290a'),
    title: 'Machine Learning Intro',
    author: 'Robert White',
    category: 'AI',
    price: 600,
    rating: 4.8,
    year: 2023
  },
  {
    _id: ObjectId('69a9b2020beeaeed997c2909'),
    title: 'Data Science Guide',
    author: 'Anna Brown',
    category: 'Data Science',
    price: 550,
    rating: 4.7,
    year: 2024
  },
  {
    _id: ObjectId('69a9b2020beeaeed997c2907'),
    title: 'JavaScript Essentials',
    author: 'John Smith',
    category: 'Programming',
    price: 450,
    rating: 4.5,
    year: 2023
  },
  {
    _id: ObjectId('69a9b2020beeaeed997c2908'),
    title: 'Python Basics',
    author: 'David Lee',
    category: 'Programming',
    price: 400,
    rating: 4.2,
    year: 2022
  },
  {
    _id: ObjectId('69a9b2020beeaeed997c290b'),
    title: 'HTML & CSS',
    author: 'Chris Green',
    category: 'Web Development',
```

VIT-AP UNIVERSITY, ANDHRA PRADESH

Lab Sheet 7: MongoDB with Fronend

Branch/ Class: B.Tech/M.Tech **Date: 05-03-2025**
Faculty Name: Prof. S.Gopikrishnan **School:** SCOPE
Student name: Ponnoju Sreeram Siddardha **Reg. no.:** 23BCE9304

```
online_book_finder> db.books.find({
| rating:{$gte:4}
| }).limit(5)
[
  {
    _id: ObjectId('69a9b2020beeaeed997c2907'),
    title: 'JavaScript Essentials',
    author: 'John Smith',
    category: 'Programming',
    price: 450,
    rating: 4.5,
    year: 2023
  },
  {
    _id: ObjectId('69a9b2020beeaeed997c2908'),
    title: 'Python Basics',
    author: 'David Lee',
    category: 'Programming',
    price: 400,
    rating: 4.2,
    year: 2022
  },
  {
    _id: ObjectId('69a9b2020beeaeed997c2909'),
    title: 'Data Science Guide',
    author: 'Anna Brown',
    category: 'Data Science',
    price: 550,
    rating: 4.7,
    year: 2024
  },
  {
    _id: ObjectId('69a9b2020beeaeed997c290a'),
    title: 'Machine Learning Intro',
    author: 'Robert White',
    category: 'AI',
    price: 600,
    rating: 4.8,
    year: 2023
  }
]
```

```
online_book_finder> db.books.find()
| .skip(5)
| .limit(5)

online_book_finder>
```